

ActInf Livestream #047.2 ~ "Enactive-Dynamic Social Cognition" & "Active

Livestream | & | 1:54:32

Hello everyone, welcome to ACT-INF live stream number 47.2.

It is August 17th, 2022, and I'll just quickly update the video.

Now it's really August 17th.

We're having our third discussion of the set of papers in active dynamic social cognition and active inference and active inference and abduction.

Welcome to the Active Inference Institute.

We're a participatory online institute that is communicating, learning, and practicing applied to active inference.

You can find us at links here.

This is a recorded and an archived live stream, so please provide feedback so we can improve our work.

All backgrounds and perspectives are welcome and we'll be following video etiquette for live streams.

At activeinference.org, you can find out more about participating in projects like the live stream and also a lot of other learning groups and projects ongoing.

We're here today in our third discussion around this dyad of papers, both from 2021, active inference and abduction and inactive dynamic social cognition and active inference.

So check out 47.0 and 47.1 for more background and to kind of get up to where we are now.

Let's go to the introductions and then we're going to take it where it goes.

So I'm Daniel.

I'm a researcher in California and I'll pass it to Dean and then to Steven.

I'm Dean.

I'm here in Calgary.

I guess I'm still a researcher, although I'm a retired one.

Basically, this is an opportunity to have one of the authors with us, which I think is fantastic and essentially sort of take this from all the things that we've set up to now and get an opportunity to sort of get through the lens of the authors.

And so that I'm really excited about that.

I'll pass it over to Steven.

Thanks, Dean.

So, yeah, I'm Steven.

I'm based in Toronto.

I work a lot with spatial meaning making, participatory theatre and dialogue approaches.

And, yeah, I'm really pleased to have the author here today, one of the authors.

And also, for me, I'm really interested in what we're doing here because of the ability

to get beyond what's often just called emergence, emergence through the mind body or emergence through some sort of loop that's being explained in an action or emergence that people just talk about when things interact and something happens.

Well, I see this work with abduction or the work with abduction as being a way to make a much more tractable way for living things that start to make things happen, be thought of and not just call everything emergence.

So, yeah, I'm I see both these two as being big kind of black boxes that we can maybe shine some light on and that will help a lot of people.

So, yeah, I pass it back.

Should I pass it to you, Daniel, first?

Well, Majid, go for it and just say anything you'd like to any context on the paper.

I'm Majid.

I'm a philosopher of science based in Ankara Middle East Technical University.

And I thank the organizer for this wonderful event and for the good and graceful discussion of the ideas and themes of the papers.

I do not know if I can add anything that has not been discussed already in the past couple of sessions.

So I think that this is well, there has been for the past few years.

So I'm the second author of one of the papers, active inference, abduction and active inference.

And the other author, Othi, is a renowned Charles Sanders Peirce schooler.

So I mainly can take credit only for the parts that are both active inference.

I'm not an expert in Peirce ideas or in semiotics for that matter.

But I also have been doubling from time to time in bio semiotics.

And I was searching for a way to be able to grow semantics in nature, to be able to grow semantics in natural war, which is very tricky because it seems that to the extent that we are engaging in natural processes, we can see mechanisms, we can see processes.

But meaning is something that emerges at some point, something that has not been there before in the nature itself, which doesn't completely sound right because, well, it seems that everything has to be come from nature at some stage, but it is not like a continuum.

So I was thinking of finding a way for an action oriented way for being able to speak about meaning and representation.

And well, well, the inactivist embodied approaches in cognitive science and philosophy of science are very well established.

No, but to be able to grow semantics, semiotics in action, we came to the conclusion that active inference may provide a nice way, a nice way that is well supported by what we know from in the sense of computational neuroscience and theoretical biology.

to be able to explain how some linguistic concepts, some semiotic concepts can be explicated in natural terms.

I don't know if any of that makes any sense to you, but it was what we were trying to do in this paper and a couple of other papers that we were composing to altering together.

Excellent.

Thanks for that context.

Dean?

So, Masheed, I got to ask you a question.

So lots of times when we're talking about semantics, we tend to see them written down on pieces of paper as an example, right?

But when you say semantics in nature, one of the questions that we've kind of danced around but not sort of spoken to directly is sort of semantics as nature, maybe semantics surrounding us and the abduction paper really helped us understand maybe the perspective that instead of us moving towards and seeing semantics or generating semantics, us being available to that information around us.

What's your sense after sort of, I don't know if you've embraced Pierce's philosophy, but after at least engaging with it, what is your sense of what semantics are as nature as opposed to sort of embedded in nature?

Well, that's a very good question and also a difficult one.

Well, look, so I actually didn't, what I was trying to speak about was grounding semantics in nature, saying that, so sometimes I also think of our environment, our informational trade-off environment in semantical terms, but some other times I think that perhaps we are reading what we think about the world into the world.

Perhaps there are no meanings in the world itself.

I think that if we want to be erotically naturalist about the world, we couldn't, so the natural world doesn't contain semiotic in the sense that, for example, artificial languages, human-made languages can contain semiotics.

That much is, I think, more or less obvious.

I think that we can treat semiotic into the nature, the same kind of semiotic that we can find in human-made languages into the nature.

I think that that's obvious too.

On the other hand, the things that are made by human beings are in a sense natural, because human beings are natural entities and they have this kind of natural capacities to be able to communicate with one another and with them.

So there should be some sort of natural basis.

It couldn't just come from nothing.

There should be some sort of natural basis for that, which is basically, I like to think of that in terms of human beings or the community of human beings, action-oriented interaction with one another and with nature.

So if you think of semantic in this sense, thinking of being in semantics or being surging in semantics or enveloped in semantics in this sense, I completely agree with you.

Because I think that even though it is not precisely as well-stated semantics that we can find in an artificial language, there is some sort of trade-off of information between human beings, between organisms, self-organizing complex system with one another, biotic self-organizing complex system with another, and with their environment, either individually or collectively.

And I think that this provides a good natural basis for also understanding semiotics in the sense of the signs that we interpret, the interpreted signs that we use in writing books or I don't know, different sources of communication, the more advanced ways of communication.

Thank you, Steven.

Thank you, Steven.

Yeah, thanks, Majeed.

I was wondering, when you're talking about these more advanced forms of communication or biological ways of having semantics, whether that's far more entropic, far more dispersive than what would happen when we think of it in our, you know,

traditional literature of words and ways that we would have ontologies and computers.

So is there kind of some other, is it a completely different process or is it like another flavour of, what's your thoughts on that?

I'm sorry, I was not quite clear about what I was trying to say.

What I was trying to say is this, look, by speaking of the advanced forms of communication, I was intent to refer to advanced form in comparison with the rudimentary natural form.

So for example, the way that we are speaking together, right, or the way that Danny is just writing down the ideas on the screen and that sort of stuff.

These are, I think that these are advanced ways of doing some sort of semantic, conducting some sort of semantic enterprise because we can communicate.

And by speaking of that in terms of advanced, I just wanted to say that it is more advanced that, I don't know, the very basic cases, for example, very basic cases that we can see in the natural world.

So for example, how the signaling processes in terms of biological, biological biological biologists, cellular biology, that sort of stuff.

So I think that I'm seeing that in terms of some sort of theory of evolution, how creatures or how self organizing systems start to communicate with their environment,

even at the level of population of cells or neurons or whatever.

So there are the rudimentary ways, but at the same time, there are more advanced ways of communication, meaningful communication, such as what we do in human societies,

whether in terms of individual or what we do in libraries, institutes, and recently in terms of human artifact interact.

So it was what I, and I think that the story, the main story, what, what persuaded us, what motivated us to look into active inference was that we wanted to find some sort of natural basis for that.

And you are completely right again.

We were thinking of that in terms of, in terms of the entropy of system, keeping the entropy of this information entropy of the system within the bones.

So for, so for, for like everything else that has been constructed around the notion of active inference and free energy principle, the core of the story is that in order to be able to maximize the chance of their survival, the

biotic self organizing systems have to be able to keep their entropy or their surprise within the bones.

And in order to be able to do that, they have to be able to find ways to optimize their moods of communication.

And I think that this provides a basis for a basis for the generation of semantics.

So in order to be able to maximize the chance of survey wall, we need to be able to get to some level of competence and reaching that level of competence seems to be impossible.

So I think that to the extent that this is a story is action oriented to the extent that it tries to grown semantics in nature or semantics in nature.

I think that this is completely faithful to the, to the pragmatists, to the Persian origins.

But then again, I'm not an expert in that field.

Yeah.

Thanks.

Like what I heard there is that active inference helps us find patterns across systems by framing them within a common lens of keeping their entropy or their surprisal bounded in engagement with their environment, which can include other entities like themselves or just a niche more generally.

And maintaining that survivable bound homeostasis and all of the variants that we've discussed in other times, it's going to have different features in cells, bodies, computer programs.

Like a computer program is a specifically constructed entity where one extra parentheses or period is going to lead to a halt.

But this continuous embodied and inactive nature of the biological engagement, the biosemiosis, which is what is pointed to in the formalisms that we saw in Hippolito and Van Ness paper.

Again, the only formalisms they brought in the paper rather than citing out to were about things changing through time continuously.

And so, yes, for sure, maintaining the bounds of computer programs in a digital and in a fragile setting is different than maintaining the bounds of survivability for like a cell or for a linguistic entity.

And they are different, but they can't even be said to be different unless we have some ways of juxtaposing and contrasting them.

So, Steven, and then anyone else?

Steven, thanks.

Steven, thanks for that clarity there, Majid.

One thing that I think also this ties to and I think having this move from semiotics into like this broader kind of unifying theory of active inference is really interesting.

And it speaks to some extent as well to the when you say advanced communication, it speaks to what we often think of in complexity work around complex and complicated.

So it's like organisms and life has this complex adaptive nonlinear processing.

And then there's these kind of simple to more and more complicated ways of engaging the niche to do things to have sort of systems of performative action using objects, using in the case of words that can also be seen as objects.

So there's this kind of parallel evolution between complicated things being manipulated and simple processes of things being manipulated and the complex sort of nonlinear abductive processes of the, that's all going on

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I don't know whether it could be translated into the technical notion of complexity, but I'm saying that's like that.

So take the individual fish in a school of fish,
or take the monkeys in a horde of monkeys,
or bears in a flock, or anything of that sort.

They have to be able to try to make sense.

They are trying to, so it seems that they are engaging in some sort of harmonious activity.

No, by speaking of more advanced form,

but in terms of the sophisticated cultural forms of communication that we use,

which usually come, especially in case of science,

sometimes formulation, notation,

They demand

because you need to always test the theories

So this kind of engagement between human beings also mandates

because it is not just on the basis of transfer of pure information that takes place.

but most of the time through manipulation of the environment,

which also engages artifacts other than other agents.

And this is how I think of the sophistication or complexity of the situation,

by thinking of that in terms of the advanced forms of cultural practices,

such as what we observe in mathematics, literature, and science,

which are still meaningful,

which are still involving a great amount of semiotic enterprise, semantic enterprise,

but we need to be able to explicate them in simple natural forms of communication,

like what we observe in the natural form,

and what we hope to be able to be explicated

in terms of minimization of free energy

and keeping the information entropy of the system within bounds.

It's a different thing to maintain the,

not better or worse, but merely different,

to maintain the survivability of a biofilm than a data center.

Yet we want to approach them in a first principles,

grounded, compatible, interoperable scientific lens.

And so that is what perhaps brings in active inference.

So Dean, and then we'll continue.

Sorry.

Yes.

Yes.

Just with regard to the last thing that Dan said,

there is another collaboration.

I mean, there's another co-authored paper

that is by me and Afti published in,

I think, the European Journal for Philosophy of Science.

And it is about the explicating methodology of science,

which also includes abduction and inference to the best explanation

and other sort of stuff on the basis of active inferences.

So yeah, you are completely right.

This is, and in many ways,

this project to be able to explicate methodology of science

at the individual and social level,

on the basis of this first principles,

given all of the skepticism that there exists

about trying to engage in first principles in life sciences,

it sounds to be a very ambitious project,

but it seems that where we are now,

and I think that this is a project worth pursuing.

But if there is time, I will come back to that later.

Excellent.

Dean?

This is awesome.

I got a question for you,

because I think you could probably answer this better than I could.

I'm still wrestling with this one.

So if I'm a person who's trying to perform

some kind of a situational analysis,

trying to determine what the functions and the purposes are

in the setting in which I exist,

I can look at that in terms of a semantic space,

but what abduction,

I think,

introduces is the idea of semantic time.

And I'm so glad that you sort of pointed to the fact

that Daniel's typing stuff out right now.

This is semantics time for him.

He's actually generating some semantics,
even as we're talking.

So I guess what I'm wondering is,

if we're going to look through the lens of active inference
as the situational analysis lens,

do we separate out what Daniel's doing when he's typing
from the nature of what we're doing right now?

Or do we see them as sort of co-compatibilities?

I don't know the answer to that.

I'm leaning in the direction that they're kind of inseparable.

They orchestrate something that doesn't happen
if we only have one or the other.

What are your thoughts?

Well, to tell the truth,

I'm not well...

Sorry, my phone was ringing

and I lost focus for a moment.

And I do apologize for that.

It was one of the reasons

that I was hesitating to join.

But, well...

Well, could you...

I do, again, apologize to ask that,

but could you elaborate a bit on the last part?

I hear that right.

But I want to get more familiar with the last part.

Sure, well, the semantic spaces stuff,

I think, is really material.

Like, you can see Daniel typing things out across that slide.

But what I think curse asks us to contemplate

is the actual semantics time,

the actual change or shift

of what's actually happening in that moment.

And so I was wondering

if we are not just looking at active inference

as the object of our attention,

but we're actually inferring what is going to happen next,

do we have to include the idea of what's happening,

not just semantically as a product

of what's typed out on the page,

but as a process.

And what does that then imply

in terms of what we can...

What we might be able to pick up quicker

in terms of our embedding

in these different contexts

that we otherwise wouldn't be able to pick up

if we only had one or the other.

It kind of starts moving us towards

this realism, anti-realism debate,

which I'm still not sure it should even be a debate,

but that's me and my relativity.

Shortly, what I heard there,

Unmute, is semantic spaces

have been extremely characterized,

not completely,

but before active inference,

long before people have done
embeddings of sentences
and meaning concepts
in semantic spaces
without a temporal aspect.

And so I heard a question about
abduction as providing
a framework for
semantic or semiotic time,
whether with the two-stroke engine
of generating and selecting hypotheses
or the action perception loop
or a few other ways.

All right, very well.

Look, very well.

I think that that's a very good question.

I put only in order to
blow on one's own trumpets.

I put the paper that we wrote in the chat box.

And I think that this is precisely about
this point,
because it is about selecting hypotheses,
the process of selecting hypotheses.

Look,
and I do not know
how to relate that directly
to induction or abduction,
but what I try to do
is to say this,
that at least
in the so-called
advanced forms of
communication or abduction
that I thought about,
what's important
is not only what we are doing
in the moment,
but what's important

or estimation

of what's going to happen

later next.

And I think that it is very important

in terms of

Charles Sanders Peirce's

logic of,

not only logic of science,

but also economics of science.

when he says that we have to

try to find

a way to invest in theories,

in hypotheses

that have a chance

to being a winner

in a long time.

So,

in a way,

time enters the process

because we have to be able

to have a long time view

on how the abductions

that we are going to construct

lead to fruitful results,

not only at the moment,

but in future,

possibly long future,

especially if we are engaging

in some sort of sophisticated

sophisticated process

such as scientific practice

or such as hypothesis construction

or hypothesis testing.

Time plays an important factor.

And what's important

about active inferences,

especially the forms

of active inferences

that could be appealed to?

So,

when we speak of active inference,
perhaps we didn't elaborate
that much

on the abduction

and active inference paper,
although we referred to that,
but in the other paper,

I think that we have been
referring to that quite a lot.

The thing is that

the kind of active inferences
that are useful

in this context

are active inferences

that are taking place

on the basis

of temporally rich models,

what Christel and others

mention as models

that have some sort of

temporal depths

or temporal richness,

which basically

allow the organism

to have a perspective,

have some insight,

not only on prediction

of free energy

or, sorry,

minimization of free energy

or minimization

of information entropy,

but also have some insight

into the efficiency

of its processes,

its practices

for minimizing free energy
in the long time.

So I think that it comes
back to the issue
of temporal depths
and temporal richness,
which is also
is related
in a very interesting way
to the issues
of consciousness
and selfhood,
another topic
that I'm interested in.

So this is the thing.

There are some
biotic self-organizing systems
that can engage
in some sort of
social activity,
and these
self-organizing systems
happen to also
have this wonderful
capacity
to form abductions
that take
the temporal aspect
into account.

At the same time,
the same group
of biotic self-organizing
systems
that are capable
of making
this kind of abductions,
inferences,
are also capable

of having selfhood
and consciousness
and self-consciousness
and that sort of stuff.

So in a way,
it comes together,
the ability
to be able
to form
this kind
of abduction
that allows
to choose
the hypothesis
that have
a better chance
to be good
hypotheses
in the long run
and the ability
to have self
or self-consciousness
and at the same time
the ability
to form
generative models
that have some
temporal depths
or temporal richness.

So I think
that temporal depths
is at the root
of that all
because it allows
us to make policies
and it allows
us to have
some strategies

some strategies
that basically
enable
what we can think
of in terms
of scientific
methodology
in terms of
philosophy of science
that I'm interested in.
If I could
add a note there
then pass back
to Dean
in thinking
about the similarities
and the contrast
with different
biosemiotic systems.
So a signaling protein
or a signaling molecule
might have
a tremendously
rich
and contextual
biosemiosis
like a hormone
in a certain way
might do
various different
things
relationship
with the cellular
context.
So
what I'm hearing
with the
more rudimentary

forms
or the non-abductive
forms
of communication
and the relationship
with their temporal depth
is that
while there is
a conditionality
to the way
that hormone
is received
and there's
an instantaneous
complexity
or complicatedness
it doesn't
necessarily have
a temporal depth
in the semiosis.
It might play
a role
in a temporal depth
like the hormone
could play a role
in development
but the interpretation
of that
hormone
is instantaneous.
In contrast
with linguistic
communication
even before
we get to this
level of
speaking to
the distant

future
or invoking
the distant
past
or different
timelines
and metacognitive
awarenesses
there's a space
of interpretation
between
the heard
sound
or the seen
symbol
itself
temporal depth
and the way
that it
triggers
internal
cascades
which must
involve temporal
depth
like a word
alone
a phoneme
alone
is not
conveying that
same impact
of meaning
as the hormone
binding
and so
the increased
cognitive complexity

of like a
linguistic
entity
opens up
a space
of interpretation
for the words
that brings
in the temporal
depth
and differences
in acculturation
and differences
in what people
have learned
whether they
even know
that word
at all
whether they
know about
how it's
being used
by that
person
and so
in the
linguistic
semiosis
we can
see how
it has
analogy
to other
forms
yet also
contains a
space where

also active
inference can
be used
to start
to disambiguate
and to
build models
of

Dean

Thanks Daniel

first of all
you almost
grabbed what
I was about
to say
but I'm
going to
say it
in different
words
here's a
question
again
around
abduction
paper
I don't
have the
exact quote
up here
but so
paraphrasing
one of the
things that
in the paper
that you
point out
is that

Pierce
understands
that if
you if
you don't
see the
triadic
if you
don't see
the three
kinds of
logic
as taken
together
the deductive
the inductive
and the
abductive
you're kind
of missing
what he
was trying
to alert
us to
my
question
is
so
when
we
break
it
down
further
from
that
if
we

reduce
it
to
deductive
inductive
and
abductive
we
don't
leave
it
as
a
whole
as
the
way
of
looking
at
all
of the
logic
sort
of
working
in
concert
!
Is
that
looking
at
logic
as
a
tool?
Is

it
looking
at
it
as
instrumentalism
if we
take it
down to
that
reductionist
place?
I know
that if
we leave
it up
at all
three
together
working
in
concert
that's
realist
and that's
the part
that I
very much
agree
with in
your
paper
but then
my
question
was
you
guys

never
really
talked
about
this
if
it's
taken
down
to
those
more
simple
relative
logics
over
time
are
each
one
of
those
logics
then
could
we
see
them
as
actually
being
instrumental
well
I'm not
sure
that I
can
answer

that
question
because
I'm
as I
say
I'm
not
an
expert
on
pairs
but
I
can
say
my
own
perspective
okay
so
what I
was
genuinely
interested
in
was
in
engaging
with
pairs
ideas
was
the part
that was
about
abduction
so

perhaps
I was
I don't
know
I know
that I
think
of
this
I do
not
know
why
I
was
so
interested
in
abduction
but
not
in
deduction
or
induction
but
perhaps
because
inference
to the
best
explanation
abductions
are
seen
do
seem
to

play
an
important
rule
in
philosophy
of
science
but
well
it
seems
that
philosophers
of
science
just
take
deduction
and
induction
for
granted
and
perhaps
it
is
because
I
can't
defend
that
decision
I don't
know
whether
I had
to

spend
more
time
to
study
deduction
and
induction
but
about
these
logics
or at
least
these
kinds
of
inferences
being
in
sterylumens
I
don't
see
why
they
couldn't
be
in
sterylumens
in
a
way
if
I
want
to
stick

to
the
jargon
of
distributed
cognition
and
embodiment
and
inactivism
the things
that I'm
pursuing
no
I think
that we
can
think of
them
in terms
of
cultural
tools
like
many
other
kinds
of
cultural
tools
that we
can
use
in
order
to
pursue
intellectual

quest
whatever
that
intellectual
quest
is
there
to
be
pursued
so
I
think
that
we
can
think
of
them
as
instruments
so
the
main
question
I
think
that
because
you
also
mentioned
realism
and
anti-realism
the
question
is

that
if
we
think
of
these
things
as
instruments
would
there
be
any
room
for
being
realist
about
this
kind
of
approach
is
that
related
to
the
question
of
realism
what
do
you
want
to
say
what
do

you
want
to
say
do
you
want
to
relate
that
to
the
question
of
realism
yeah
actually
do
I
think
as
I
said
when I
read
the
paper
I
think
the
argument
that
says
that
if
we
look
at

Pierce's
work
as
anti
non-nominalism
non-nominalism
I believe
when it's
in its
complexity
state
and
the
relationship
between
those
logics
it's
the
relationship
between
those
logics
that
it's
examined
yes
I
very
much
agree
you
can't
describe
that
as
anything
other

than
realist
my
question
is
is
that
if
we
reduce
below
that
is
that
when
the
instrumentalism
percolates
that's
my
question
all
right
very well
look
so
this
is
my
perspective
on
realism
in
this
context
is
this
look

so
we
are
speaking
about
semiotics
and
we
are
speaking
about
the
semantic
aspects
of
language
and
interpretation
and that
sort of
stuff
the question
is that
where this
kind of
interpretation
where this
kind of
semantical
interaction
with
the world
is coming
from
is
I
think
that

if
we
can
see
that
it
is
not
something
arbitrary
that
this
is
something
that
is
not
developed
or
chosen
on the
basis
of
convention
not only
conventions
but conventions
that are
feeding on
I don't
know
and
whimsical
element
if
there
is
some

objective
element
in
that
it
is
not
instrumentalism
it
will
provide
some
basis
for
realism
!
So
and
there
are
semiotic
and
semantic
aspects
in
for
example
constructing
scientific
hypotheses
or
theories
or
scientific
practice
and
that
sort

of
stuff
right
I
think
that
to
the
extent
that
we
are
able
to
grow
on
these
semantic
and
semiotic
aspects
in
nature
to
the
extent
that
we
can
provide
some
sort
of
natural
explanation
for
the
emergence

of
this
semantic
and
semiotic
element
we
could
defend
some
sort
of
realism
realism
so
then
again
you
may
ask
what
sort
of
realism
is
that
and
I
can
say
that
this
is
a
sort
of
realism
that

is
informed
by
our best
scientific
theories
so
it
is
some
sort
of
radically
naturalist
take
on
realism
that
I
think
would
be
kind
of
compatible
with
pragmatist
and
Persian
ideas
so
I
think
that
the
realist
take
comes

from
the fact
that
we
try
to
grow
on
semiotic
and
semantic
the
logic
of
abduction
induction
deduction
all of
that
we
try
to
find
natural
processes
that
enable
those
kind
of
inferences
and
to the
extent
that
we
can
find

natural
processes
that
are
explicated
in
terms
of
some
of
best
scientific
theories
there
remains
some
room
for
defending
a
version
of
realism
that
Willard
Van
Orman
Kuin
used
to
call
robust
realism

It
depends
instead
of

trying
to
latch
onto
mind
independent
science
independent
realities
into
the
objective
world
this
kind
of
realism
draws
its
force
from
or
best
scientific
theories
and
I
think
it
is
realist
in
the
sense
that
we
can
provide

a
naturalist
story
we
can
flesh
out
some
natural
natural
natural
story
some
scientific
story
of
the
psychological
grown
mechanisms
that
enable
this
kind
of
inferences
awesome
thank you
thank you
Steven
and
then
I'll
add
a
note
thank
you

when
you
mention
so
you
mentioned
the
scientific
process
and
almost
the
deductions
the
inductions
that can
occur
with
a
model
a
scientific
model
that's
being
created
so
it's
slightly
different
to
say
the
what's
real
for
me
at

the
scale
at
which
I
can
perceive
as
a
human
and
see
things
my
model
might
be
able
to
see
things
slightly
beyond
what
my
sensorium
can
perceive
in
my
experiential
ecosystem
I like
to
use
the
term
experiential

ecosystem

of

being

alive

so

when

that's

the

case

is

it

almost

that

these

models

are

given

some

sort

of

agency

some

sort

of

sentience

that

what's

real

for

the

model

in

a

way

all

right

very

well

that's
a
very
good
question
in
order
to
try
my
best
to
provide
an
answer
to
that
I
may
need
to
drift
away
a
bit
from
the
scientific
context
and
engage
in
some
sorts
of
philosophical
discussions
that

might
be
a
bit
boring
for
for
the
audience
I'm
not
sure
about
that
but
I
will
ask
you
to
try
try
all
right
well
giving
some
sort
of
agency
to
models
is
an
interesting
question
a question
that

as it
happens
I think
is worth
taking
seriously
in
but
it
might
not
be
directly
related
to
look
in
order
to
be
able
to
defend
some
sort
of
in
order
to
advocate
some
sort
of
model
based
version
of
realism

because

you

do

not

necessarily

need

to

give

agency

to

models

and

because

in

order

to

be

realist

you

the

thing

is

this

the

nature

and

structure

of

scientific

theories

is so

complicated

that

we

cannot

defend

realism

on
the
basis
of
some
sort
of
perceptual
or
sense
perception
evidence
so
the
evidence
for
scientific
theories
if
you
think
of
that
in
terms
of
model
based
science
doesn't
come
in
terms
of
sense
perception
because
the

construction
of
scientific
theories
is
much
more
complicated
than
that
there
are
different
kinds
of
experimentation
manipulation
theoretical
distortion
idealization
you name
it
so
when we
speak
of
models
there
are
different
kinds
of
things
that
make
it
very
hard

to
think
of
the
model
based
representation
as
some
sort
of
direct
relation
so
we
cannot
defend
realism
about
models
in
terms
of
relating
sense
perception
or
sense
data
or
something
directly
to
the
model
there
are
other

kinds
of
virtues
that
we
may
want
to
depend
on
and
most
of
the
time
philosophers
of
science
do
not
agree
on
a
specific
set
of
virtues
that
could
be
used
they
usually
speak
of
considerations
of
simplicity

fruitfulness
personally
that
sort
of
stuff
efficiency
but
usually
they
do
not
agree
on
a
specific
set
of
virtues
but
one
thing
that
they
agree
about
is
that
models
do
not
receive
confirmation
on

the
basis
of

their
feet
with
sense
experiences
because
it
seems
that
our
modern
science
or
modern
think
of
that
in
terms
of
quantum
mechanic
things
not
only
that
but
different
sorts
of
sciences
is
becoming
so
theoretically
complicated
or
scientific

models
are
becoming
so
idealized
and
theoretical
entities
that
feature
in
these
models
are
sometimes
so
sophisticated
that
it
would
be
futile
to
try
to
find
some
sort
of
perceptual
or
sensory
evidence
for
that
right
so
well

and
again
when we
speak
of
abduction
and
when we
speak
of
inference
to
best
explanation
I
still
think
that
active
inference
could
eventually
or
could
at
some
point
lead
to
at
least
grow
some
methodology
of
model
selection
that

allow
us
to
on
the
same
basis
on
the
more
or less
on the
same
basis
that
Charles
Sanders
person
speaks
of
logic
of
science
or
economics
of
science
can
lead
to
some
sorts
of
methodology
of
science
that
allow

that
that
could
allow
us
to
select
scientifically
fruitful
models
models
that
deserve
to
be
given
a
chance
by
realists
as
things
that
are
providing
a
more
or
less
relatable
image
of
the
universe
without
depending
directly
on

sense
experiences
Thanks
there's
a lot
there
one
of
the
most
key
pieces
of
I think
the
juxtaposition
of these
papers
and
how
we're
seeing
the
discussion
around
science
and
realism
and
enculturation
progress
in
2022
is that
and to
connect
this
also

to
this
area
of
the
semantic
spaces
and
semantic
time
which
abduction
requires
it's at
least
two
phasic
so it
has to
it can't
both be
happening
in the
same
instant
in the
same
unit
so
the
question
of
realism
and
especially
on
specific
constructs

within
the
broader
apparatus
of
free
energy
principle
like
focusing
on
the
blanket
construct
Markov
blanket
Friston
blanket
etc
it's
often
been
approached
in
terms
of
a
question
of
composition
of
space
as
you
put
it
like
mind

independent

does

this

exist

physically

well

what

do

you

mean

by

the

table

exists

physically

I

mean

it's

there

when

I'm

not

looking

or

if

I

don't

believe

it's

there

it

could

still

hit

me

in

the

mind

independent
situation
first off
there's a
massive
challenge
in
re
minding
that
system
the
whole
hard
problem
of
consciousness
or
even
just
simply
hard
problem
of
biocognition
or
semiosis
and so
in that
setting
it's
very
defensible
to
take
a
instrumental
perspective

on
the
meaning
of
semantic
spaces
when we
turn from
a
compositionality
of
mind
independent
physical
space
into a
compositionality
of
meaning
as
enacted
in the
pragmatic
sense
what's
real
is
action
then
the
commitment
to mind
independence
being lifted
we
have
actually
a

multi
perspective
cognitive
scenario
that
naturally
embodies
questions
of
semiosis
because
meaning
as
enacted
is
model
dependent
and so
I'm sure
there's a lot
of like
loose threads
or other
ways to
go
but
again
it's a
large
turn
for me
to see
the
discussion
evolve
from
being
about

mind
independent
physical
interpretations
as being
the crux
of realism
and the
pragmatic
shift
as opposed
to like
the
materialist
or the
reductionist
stance
and the
pragmatic
shift
reintroduces
mind
or really
just
removes
the kind
of filter
or the
dam
and just
lets mind
rush back
in
so there's
really a lot
there
and then
you connect

that all
the way
to
active
inference
grounding
methodologies
of model
selection
and I
think in
that
domain
it
inherits
greatly
on
a huge
amount
of statistics
and machine
learning work
on policy
selection
for example
like using
a free
energy
functional
and minimizing
it
to
scan
and evaluate
and make
a decision
upon a space
of affordances

and policies
is what
has been
happening
in control
theory
cybernetics
and so on
even down
to the
evidence
lower bound
and the
forms of
the free
energy
functionals
that are
used
are broadly
related to
exactly
model
selection
processes
so just
like if we
were doing
a linear
regression
we might
want to
fit
the least
squares
error
and if
we were

doing a
Bayesian
model
selection
we might
want to
use like
a base
factor
or a
BIC
or AIC
criterion
here we're
in the
space of
mind
based
cognitive
entities
active
particles
and so
active
inference
provides
a process
theory
that
is able
to
make
all of
that
come
together
and
make

sense
and
then
this
paper
connected
it
to
the
qualitative
though
rigorous
and
structured
framework
of
abduction
and
kind
of
fit
like
abduction
into
the
ignition
instead
of the
inductive
deductive
which can
be
limiting
in
certain
contexts
Stephen
just

just
wanted to
pick up
back on
when you
mentioned
about
model
selection
so
that's
interesting
in the
sense
that
there's
there's
policy
selection
in active
inference
okay
what policy
I'm going
to take
on my
action
I suppose
you can
look at
what
regime
of
attention
will I
use
and will
capitulate

my
internal
model
but I
think a
lot of
times
and I
think
this is
often
thought
about
in
active
inference
but
really
the
model
selection
may
well
be
the
external
artifact
or
the
like
Majee
was talking
about
the
scientific
model
is
different

to say
the
model
selection
that's
happening
in terms
of maybe
types
of
generative
models
that we
might
think
of
in
the
brain
and
can
have
its
own
you
know
it
can
exist
someone
could
write a
paper
like
Pierce
and
you know
it wasn't

published
and people
pick it up
50 years
later
and it's
recapitulated
into

a
system
of
meaning
making
so
I
think
there's
the
ways
that
models
get
selected
and
then
how
they
can
be
used
and
maybe
couched
in a
system
I think

is really
interesting
and maybe
that
system
is also
we've
talked
about
this
before
when
that
system
doesn't
need
to be
alive
anymore
to
function
like
can
there
be
a
dead
system
i.e.
a computer
system
that can
then
or how
much of
it
becomes
not

dependent

on

the

living

so

that's

just

one

point

that

I

think

comes

back

to

that

question

of

where

is

it

ultimately

grounded

Thank

you

Stephen

yes

Majid

I

think

that

unfortunately

I

have

to

start

to

leave

and
I
do
not
want
to
open
another
can
of
warmth
before
doing
that
but
I
want
to
briefly
speak
about
what
Stephen
mentioned
right
for
example
and
I
want
to
say
with
the
example
that
you
provided

for
example
the
person
being
dead
Charles
Sanders
being
dead
but
the
ideas
of
being
picked
up
when
I
was
starting
to
think
of
scientific
practice
in
terms
of
active
inference
in
a book
that
I
wrote
in
2019

people
started
to
make
very
significant
criticism
and
meaningful
that
active
inference
is a
theory
of
individual
brains
but
scientific
practice
is not
an
individual
activity
it
is
not
something
that
science
is
not
taking
place
in
the
brain
of

individual
people
it
is
some
sort
of
social
activity
and
it
is
distributed
in
scientific
groups
and
scientific
communities
so
and
then
I get
more
deeply
engaged
with this
literature
on
distributed
scientific
cognition
in terms
of works
of
Ronald
Gere
and

some
other
people
and
I
found
it
very
interesting
so
look
the
model
selection
scientific
model
selection
as some
sort of
advanced
form
of
cognition
finis
form
of
cognition
is
grounded
I still
think
that it's
grounded
in
active
inference
but
it

doesn't
need
to
take
place
at
the
level
of
individual
scientists
it
could
be
distributed
to
the
community
of
scientists
and
then
the
issue
of
minimization
of
entropy
could
be
understood
in
terms
of
minimization
of
collective
entropy

of
scientists
or
even
minimization
of
the
collective
entropy
of
the
couple
community
of
scientists
and
their
environment
I think
that it
leads
to
some
sort
of
revolutionary
understanding
of
the
nature
of
scientific
theories
because
we
usually
think
of

scientific
theories
as
a
class
of
propositions
or
set
of
model
theoretic
or
set
theoretic
models
in terms
of
mathematics
but
if
we
broaden
the
scopes
a bit
to
think
of
theories
as
pieces
of
information
pieces
of
knowledge
that

are
dispersed
and
distributed
in
scientific
community
or
things
that
emerge
through
the
interaction
between
scientific
community
or
scientific
group
and
their
environment
I think
it may
lead to
a very
interesting
perspective
on the
nature
of
scientific
theories
as
well
so
in

a
way
I
was
hijacking
this
discussion
of
active
inference
because
being
trained
and bred
as a
philosopher
of
science
I'm
mainly
interested
in
some
internal
issues
in
philosophy
of
science
and
when
I
speak
of
scientific
realism
I
think

of
that
in
classical
terms
in
philosophy
of
science
being
realist
about
the
target
system
of
theories
and
scientific
models
but I
think
that
it
still
leaves
ample
room
for
also
being
realist
about
active
inference
itself
but for
different

reasons
and for
reasons
that are
much more
sophisticated
and complicated
that could
be unpacked
directly with
relation to
mind
independent
reality of
active
inferences
or whether
there are
such things as
first principles
in mind
independent
world
so I
think
that
of
course
there
are
no
such
things
as
first
principles
mathematical
first

principles
in
if you
open up
brain and
nervous system
you can't
find
first
principle
mathematical
first
principle
you will
find
different
kinds
of
tissues
and
mechanisms
and
other
kinds
of
things
but
I
think
that
these
two
concern
about
realism
about
active
inference

and
free
energy
principle
and
my
own
concern
about
scientific
realism
about
targets
of
scientific
theories
that
could
be
selected
on
the
basis
of
active
inference
could
be
related
together
at
some
point
but
I
have
to
apologize

this
is
a joy
to
participate
in
this
discussion
and
I
have
to
apologize
for
for
being
pushed
to
leave
abruptly
it's
excellent
Majid
we
appreciate
it
and
the
dot
three
is
yours
when
the
time
is
right
thank

you
very
much
it
was
very
nice
to
meet
all
of
you
I'm
going
to
leave
yeah
take
care
wow
okay
we'll
take
a deep
breath
and then
we'll
continue
or
maybe
you
don't
even
need
to
breathe
Dean
go

for
it
well
I
just
love
the
fact
that
the
term
hijacking
was introduced
not by
me
because
one of
my
biggest
fears
around
this
dot
two
was
hijacking
something
and
I
think
I
understand
what
he
means
by
that
because

it
actually
feels
that
way
if
you
look
at
what
we
were
just
talking
about
which
is
active
inference
could
ground
a
methodology
of
model
selection
let's
just
assume
for a
second
abductively
that
it
does
it
can
and

it
does
well
what
would
that
mean
well
what
that
would
mean
is
is
literally
changing
the
directionality
of
what
we
typically
perceive
and
as
as
you
pointed
out
we
can't
let
our
perceptions
conflate
what
we
are

perceiving
with
whatever
reality
is
and I
think
we've
all
agreed
that
that's
dangerous
thing
to do
however
in the
spirit
of
what
could
active
inference
ground
in terms
of
methodology
and then
in terms
of
model
selection
my
question
is
typically
typically
typically

in the
outside
world
as
we're
using
scientific
principles
we
start
with
an
idea
a
hypothesis
a
process
and
we
turn
that
into
a
product
some
kind
of
materialization
Stephen
and I
were actually
chatting
about this
before we
came on
the
live
stream

today
that's
what
we
see
that's
what
we
sometimes
label
the
reality
what
active
inference
as
active
inferring
suggests
is
the
Markov
blanket
stays
in the
middle
between
the
generative
model
and
the
generative
process
however
if we
were to
actually

contemplate
that
active
inference
could
ground
a
methodology
and
does
we
would
actually
say
that
the
generative
model
is
the
product
or
is
the
materialization
and
as
we
pass
through
the
Markov
blanket
we
actually
move
to
two

different
versions
of
process
one
is
relativity
the
other
is
relationship
and
together
that
forms
a
new
absolutism
so
that
is
literally
the
emergence
and
that's
kind
of
where
I
started
in
this
I
haven't
sort
of
been

that
explicit
on
my
backside
out
there
to
have
a
whole
bunch
of
people
take
a
bite
out
of
it
but
if
we
I
think
are
courageous
enough
and
take
the
risk
and
we
look
at
that
differently

than
how
we
traditionally
looked
at
it
that
process
leads
to
product
another
way
of
describing
that
is
product
and
process
but
we
actually
look
at
it
as
process
in
product
meaning
we
start
with
whatever
our
hard

evidence
is
as
a
generative
model
as
product
and
then
from
that
we
gain
insights
and
intangibles
and
things
that
we
call
structure
because
we
understand
now
the
relationship
and the
relativity
and
those
new
absolutisms
that
would
be

the
condition
under
which
active
inference
actually
does
ground
model
selection
what do
you guys
think
about
that
it's
a
nice
angle
I'd
like
to
move
to
the
images
that
you've
prepared
and
see
where
that
comes
back
so
let's

return
to
the
initial
juxtaposition
slide
that we
looked at
in the
dot
zero
with
the
iceberg
okay
so
again
I'll
put
our
video
chat
as
the
tip
of
the
iceberg
we
raised
many
points
of
contact
between
these
two
papers

and
then
you've
provided
a graphical
abstract
we could
say
that we're
going to
go through
now
and
then
from
that
graphical
abstract
we're
going to
enter
into
one or
more
of
these
comparisons
and
see
how
they
are
structured
in
relationship
with
the
general

abstract
which
we'll
see
now
so
here
we
are
what
is
happening
on
this
slide
okay
so
Inez
recently
had
there was
a YouTube
clip
where
she was
talking
I think
the topic
of the
conversation
was
the
free
energy
principle
and
the
edge

of
chaos
and
one
of
the
quotes
that
I
pulled
was
she
made
the
statement
making
this
bridge
logically
sound
because
what
she
was
looking
at
was
some
of
the
debate
that's
been
going
on
between
realism
and

antirealism

and

of course

in order

to

be

able

to

talk

about

those

things

she

spent

a lot

of

time

looking

at

free

energy

principle

and

how

that

works

and

she

did

a

really

brilliant

job

of

trying

to

make

a
case
for
why
the
debate
was
taking
place
why
she
and
some
of
the
authors
that
she
works
with
have
taken
the
position
that
they've
taken
what I
think
that
us
bringing
two
topics
like
that
that's
been

focused
on
inaction
and
social
cognition
and
abduction
if
we bring
those
things
together
approximately
which
is
what
we've
done
in
this
live
stream
series
we
could
actually
show
how
potentially
you
can
move
from
a place
that says
well
if

I'm
all about
the
realism
then
I can't
be
anti-realist
and vice
versa
I
think
what
if
you're
in
a
position
where
you're
trying
to
help
people
understand
not
what
they
see
outside
of
themselves
as
a
product
going
through
an

assembly
line
like
somebody
who's
learning
and
then
they
come
out
the
other
end
and
they're
now
all
assembled
because
they've
gone
through
a
process
if
you
actually
start
out
as
seeing
this
human
living
system
as
a

product
that
is
made
up
of
things
that
are
already
present
priors
and
posteriors
and
all
of
these
actions
that
it
can
take
as
an
agent
and
you
actually
move
from
the
left
of
that
bottom
left
corner

diagram
to
the
right
you
can
actually
use
the
inaction
and
the
abduction
and
the
anti
realism
and
the
realism
as
placeholders
within
a
Markov
blanket
setting
and
as
you
move
through
time
and
as
I
quote
passing

through
time
as
in
depth
the
field
some
of
those
memory
traces
that
you
are
pushing
through
your
working
memory
and
pulling
out
from
your
long
term
memory
as
well
as
literally
the
river
of
time
flowing
you

could
move
from
an
idea
of
this
is
the
platform
product
or
stability
the
ground
grounding
from
which
I
start
and
as
I
become
more
and
more
familiar
with
what
I'm
analyzing
situationally
I
can
actually
pick
up

the
relationships
the
entailments
and
the
relationships
as
relativity
and
the
relationships
as
entanglements
and
so
that
is
acting
as
an
active
inferred
as
opposed
to
looking
at
and
through
active
inference
lens
so
that's
what I
was
trying

to
pull
together
here
because
as
it
says
in
the
middle
of
this
slide
as
active
infers
as
prediction
area
reducers
across
all
processes
not
just
subject
matter
one
of
them
could
be
active
inference
as
a
tool

or
as
a
non-tool
when
is
active
and
this
was
we've
already
talked
about
the
whenness
of
semiotics
when
is
active
inference
both
a
partitioner
and
a
framer
and
when
is
it
also
a
filter
so
looking
back

through
time
and
then
pulling
from
the
memorial
and
projecting
out
into
the
future
how
do
we
keep
the
complexity
how
do
we
keep
the
triadics
how
do
we
avoid
the
trap
of
reducing
down
but
then
not

being
able
to
build
back
up
to
the
complexity
scale
where
the
reality
actually
exists
because
I
think
if you
cross a
certain
threshold
and you
get to
the
operations
level
of course
it's
got to
be
tools
of course
it's
got to
be
instrumentalism
but

then
it's
got
to
come
back
to
and
I
think
Daniel
you
mentioned
this
earlier
it's
got
to
come
back
to
so
what
about
that
table
that
I
bump
into
is
that
not
real
and so
I think
it's
the ability

to sort
of go
below the
threshold
and then
resurface back
above
it's being able
to understand
what the
actual outer
boundaries are
and what the
rows and columns
and their ability
to differentiate
are
and I think
if we wanted
to
as I said
I don't want
to hijack
but if we
wanted to
we could
show how
both
the
inaction
and the
logic
application
requires us
or at a
minimum
at least
gives us

the complexity

state

from which

active

inference

which was

it's kind

of the

one thread

that holds

all of

this

together

retains

that sense

of okay

it's both

anti-real

and real

at the

same

time

thank you

dean

stephen

first

yeah

deep

breath

so

yeah

no

this is

really

this is

there's a lot

here

and this

is really
powerful
and I
think
I think
the idea
of moving
through
these
ways
of
interpreting
anti-real
realist
so
moving
through
or moving
through
inaction
abduction
I think
that's
helpful
and the
idea
of
how that
passing
through
happens
what we're
trying to
do is
pass
through
almost
the

trough
a bit
between
the
two
and there's
this idea
that there's
maybe a
couple of
roads
we traditionally
have this
high road
low road
but it
can be
I see
it's also
the way
that we
structure
that
understanding
of
abduction
of
inaction
of
realism
is
shown up
in our
idioms
it shows up
in the way
we structure
our language

so to what
extent if we're
passing through
the middle
here
right through
the middle
and if I
was to go
and on my
left there's
a road
on my
right there's
a road
and the
road that's
left to
be explored
or the
road that's
forever
giving and
revealing
the left
field is
the inaction
the abduction
and what's
right is
my
anti-real
real
I mean it's
either anti-real
or it's real
right
that's right

as they say
in North America
a lot so I
think there's
this way
that we're
sort of
trying to
engage with
these ideas
as practitioners
and researchers
and
kind of
interdisciplinary
sort of
connectors
which starts
to become
itself
structured
so I think
this is
this diagram
is quite
helpful
to start
to tease
out some
of that
in again
maybe a
kind of
an inactive
way
thanks
I'll make
a few

comments
and then
let's
jump into
the
comparisons
and see
how this
image is
applied
so
the epistemic
foraging
the journey
the learning
journey
all of these
have a
temporality
and so
just wanted to
quickly note
that connects
very much
to our
discussion
on semantic
space
and semantic
time
okay
if there's
a space
to be
semantically
exploring
but we're
taking a

path in
it
then
first off
Dalton and
Maxwell et al's
path formulation
of active
inference is a
natural fit
and we
need to
consider
the
when-ness
and as
we've
talked about
when
there's
clock time
with
chronos
and then
there's
the
timeliness
with
kairos
and so
that epistemic
journey
we don't
need to
deny
or even
take a
realism

or not
stance
on the
chronology
we do
need to
take a
reality
of that
journey
in the
abductive
semiotic
space
with time
then
Stephen you
brought in
this road
and path
metaphor
and so
one thought
there was
these are
like signposts
like they're
labeling the
quadrant
but we
kind of have
a cultural
norm that
actually that
quadrant is
all labeled
it's not
only labeled

underneath
the word
and
so
when we
have a
signpost
and we
say we're
going along
this road
that doesn't
mean you
just gravitate
and collide
with the
signposts
in fact
we're always
like moving
between and
among signposts
and then I
just thought of
someone saying
like I'm
in the city
I'm as
close as
you can
be to
the one
pin
on google
maps
or something
like that
but actually

even these
labelings
have a
diffusive
aspect
to them
so
outside
of
important
and
clarifying
scholarly
debates
where we
are
like with
both feet
on one
label
and trying
to knock
others off
their perch
with both
feet on
their label
which we
looked at
in the
sort of
survivor
context
previously
the people
in the
audience
and in

the crowd
and the
co-creators
that the
Hippolito
and Venice
paper really
artfully
focuses attention
on
it's like
what about
the fact
that social
cognition
is distributed
and inactive
and in
culture
what about
that
they're
in that
trough
between
the
signposts
so
those are
a few
aspects
and then
one last
one was
like
Dean
you mentioned
relation

as
relativity
and relation
as
entanglement
and it
makes me
think about
as this
legend
helps us
see
aspects
of
edges
can mean
different
things
we look
at a lot
of base
graph
representations
where edges
have a
certain
interpretation
in a
statistical
framework
but even
then
with
effective
functional
and
anatomic
connectivity

we've also
talked about
different kinds
of edges
and now
here
we're in
the space
time
of
semiosis
and there
are edges
and some
of those
edges
reflect
relationships
amongst
nodes
and others
are entanglements
so
the
ankle
bone
and
the
knee
bone
might
be
like
entangled
in
the
physical
real

world
whether
we
know
about
that
link
or
not
there's
something
to it
that's
real
in the
sense
that
Majid
is
using
realism
and
then
there's
also
a
relationship
with
a
creative
enough
forger
amongst
everything
because
you can
say
the

relationship
between
the
ankle
and
the
field
of
astronomy
is
they
both
start
with
letter
A
or
you
could
it's
general
it's
not
always
a
useful
statement
to
make
well
forks
are
not
spoons
okay
but
utility
is

just
one
of
the
virtues
or
the
criteria
by
which
we
might
evaluate
a
certain
epistemic
path
and
our
harshness
on
grading
or
evaluating
epistemic
paths
doesn't
always
have
to
be
extreme
and
there
doesn't
have
to
be

one
rubric
either
so
there's
so
many
connections
in
that
space
pragmatism
and
the
productivity
and
the
social
setting
where
all
productivity
is
ultimately
embodied
brings
pieces
in
that
are
hopefully
of
epistemic
and
pragmatic
value
let's
go to

the
comparisons
or
do
you
want
to
stay
here
Dean
no
I
just
want
to
point
something
because
in
order
for
the
comparisons
to
make
sense
I
have
to
kind
of
context
so
these
arrows
I
want
to

just
say
quickly
the
interdirectionality
results
from
limits
or
constraints
things
we've
used
expressions
upper
and lower
bounds
and
bi-directionality
result
from
parallel
processing
things like
arbitrary
parameterization
which
Blue
and I
and you
talked
about
a lot
last
week
the
arbitrariness
of that

then
then
the
question
becomes
does
a
bridge
right
as
an
idea
remain
a
bridge
as
a
materialization
as
entailments
things like
the
relatives
part
that
Carl
talks
about
the
parents
and
the
children
and
we've
been
talking
about

this
a lot
and
entanglements
meaning
the
relationship
of
same
and
different
so
how
does
that
play
out
typically
from
an
academic
way
of
producing
that
is
to
start
with
what
we've
got
to
come
up
with
a
process

we've
got
to
come
up
with
a
methodology
to
derive
a
product
what
I'm
suggesting
active
inference
does
is
it
starts
with
a
whole
bunch
of
products
it's
like
what
it
is
and
and
Thomas
talked
about
in

their
paper
you
co-construct
that
social
space
you
start
with
a
product
and
then
you
learn
how
to
translate
and
interpret
one
other
quick
thing
on
the
other
side
of
this
paper
it
says
a
markup
blanket
is

a
scale
free
statistical
tool
yes
yes
yes
it
is
and
in
Majid
and
Adi's
paper
in
order
to
stay
in a
non-equilibrium
city
stay
with
its
environment
an
organism
must
regulate
its
internal
entropy
the
markup
blanket
of

a
node
consists
of
its
parents
children
and
co-parents
that's
all
true
so
it
depends
on
whether
you
start
with
a
product
and
move
to
the
process
or
whether
you
start
with
the
process
and
move
to
the

product
and
that's
where
things
really
open
up
and
optionality
really
becomes
a
thing
with
active
inference
or
at
least
that's
what
I'm
trying
to
argue
also
in
that
I
saw
a
bridge
not
as
just
the
thing

that
you
drive
on
but
a
bridge
is
a
stance
it's
a
position
like
a
backbend
and
it's
a
game
and
in
in
that
interpretive
space
is
the
difference
between
the
signaling
molecule
at
the
cellular
level
and

human
linguistic
communication
so
not to
return
only to
that
but
in
that
space
we
can
respect
whatever
led
that
generative
model
of
NS
to say
that
and mean
that
and
it's
always
going to
be a
yes
and
with
our
interpretation
and
to

the
extent
that
there
are
anti-virtues
of
model
comparison
speaking
for
another
cognitive
entity
could be
something
like
that
because
you're
never
going to
escape
the
fact
that
it's
your
framing
still
of
their
intentionality
so
how are
we
going
to

take
that
as
a
starting
point
rather
than
shut
the
door
on
discussion
because
of
that
being
a
real
feature
of
social
discourse
Stephen
yeah
and this
sort of
further to
that
product
process
and how
that fits
and
potential
bridges
that gives
Dean

can I
just ask
you
are you
saying
as well
this
this
links
to the
idea
of
the
model
in
again
we're
coming
back
to
the
idea
of
modeling
rather
than
the
model
so
with
active
inference
you
know
you
you're
you're
not

you're
not
wheeling
in
the
trojan
horse
with
the
model
that
you're
going
to
take
out
you're
you're

you're
bringing
in
a
modeling
process
where
the
product
can
be
retro
interpreted
almost
or
reverse
engineered
or
somehow

so
then
maybe
there's
something
in
that
would
that
make
sense
to
what
your
your
tool
yeah
so
again
it's
easier
to put
down
a
critical
path
because
people
are
familiar
with
that
right
like
this
is
the
things

these
are
the
steps
check
these
boxes
and
this
will
be
your
result
all
this
is
doing
is
showing
the
inversion
of
that
and
what
the
possibility
is
so
I'm
agreeing
with
you
Steven
how
do
we
bring

both
into
people's
front
of
mind
attentional
space
because
I
think
it's
both
that
are
dancing
with
one
another
as
we
are
figuring
something
out
if
we
want
to
follow
that
doesn't
require
both
but
if
we
want

to
forage
or
stigmatically
arrive
at a
place
we don't
know
other than
as an
abstraction
even
the
I
don't
want
Daniel
you're
the
expert
at
this
but
I
don't
know
what
the
abstraction
is
for
an
ant
that
has
faith
that

whatever
they
are
doing
in terms
of
following
that
trail
is
what
they
were
put
on
this
good
earth
to
do
the
puns
will
come
but I
might
even
say
that
that's
how
ants
are
let's
go to
the
first
comparison

okay
okay
we're
talking
about
yeah
go for
just
describe
the
thing
he
does
is
remove
the
limits
if
you
notice
all
four
of
the
red
lines
on
the
outside
of
the
previous
diagram
have
been
removed
because
this

was
the
inactive
social
cognition
paper
and
essentially
what
this
does
is
divide
up
that
space
it
takes
the
generative
model
and
allows
it
to
now
be
categorized
if
that's
what
you
want
to
do
if
you
want

to
move
from
the
triadic
of
abductive
deductive
and
inductive
this
is
how
you
actually
create
those
different
subcategories
within
the
larger
picture
so
there's
a
quote
from
their
paper
it's
possible
to
understand
active
inference
in
two

ways
a
realist
view
that
the
properties
of
the
model
constructed
by
applying
active
inference
tools
should
be
also
be
expected
to
exist
sorry
as
an
ontological
property
in
the
scientific
phenomenon
we're
trying
to
explain
a
non

realist
view
that
the
model
use
to
some
activity
allows
the
scientists
to
draw
interpretations
and
explanations
and
then
they
go
on
to
talk
about
why
they
use
dynamical
systems
theory
so
we
already
talked
today
about
so

what
happens
when
you
cross
a
threshold
is
that
when
complexity
arises
versus
when
you're
below
the
threshold
and
that's
when
something
different
than
complexity
exists
and
so
my
argument
would
be
yes
you
can
have
something
that's

complexity
and
then
you
can
have
something
that's
complicated
and
tools
and
instruments
are
still
complicated
don't get
me
wrong
there's
a huge
technical
investment
that you
have to
make
in terms
of being
able to
make
them
function
but
the
note
that
I
wanted

to
put
down
here
was
is
causal
order
being
explicated
not as
some
generality
as in
I'm
being
imaginal
now
on
some
abstracted
level
but
rather
as
a
specific
place
in
time
the
relational
causality
patterns
from
which
context
and

longevity
and
I
love
Majid
without
any
prompting
talking
about
temporal
depth
we
started
down
that
path
two
live
streams
ago
that
can
be
gapped
can
actually
be
addressed
as
placed
in
that
in
that
distribution
right
because

we
respect
both
difference
and
distance
within
the
collective
as
a
new
scale
as
relationally
causal
now
actively
inferring
which
includes
selection
from
within
a
distribution
the
counterfactual
which
if
this
goes
right
back
to
what
Majid
said

about
we
could
use
it
as
a
grounding
for
the
model
selection
is
not
only
able
to
function
as
a
tool
but
of
selection
by
the
selector
processing
the
action
taker
with
the
real
realm
by
the
free

energy
principle
so
we
must
first
deconstruct
meaning
if
there's
anything
within
this
that
needs
categorization
and
separation
we
can
do
that
in
order
to
co-construct
or
choose
choose
how
to
move
from
a
product
to
this
is

entailing
that
or
this
is
entangled
with
that
all
we
have
to
do
is
see
it
without
the
bounds
or
the
horizons
same
image
one
less
thing
let me
contrast
that
with
two
slightly
more
technical
notes
the
first

is
figure
9.1
in
the
March
2022
active
inference
textbook
this
is
what's
described
as
the
meta
bayesian
perspective
and
so
here
the
outer
shell
is
the
experimenter
and
the
experimental
context
that
they're
providing
in terms
of
stimuli

and
then
the
behavior
that
they're
observing
of
the
entity
in
system
and
they
are
making
a
cognitive
model
of
an
entity
so
this
is
cognitive
modeling
as
a
process
that's
one
aspect
and
it
can
be
an

active
inference
process
from
the
experimenter
without
falling
into
the
all
minds
are
scientists
fallacy
that
Yella
Brunberg
has
adequately
sketched

for
us
even
in
2016
but
we
can
still
see
an
active
inference
framing
of
the

outer
experimental
loop
and
use
compatible
frameworks
for
modeling
the
inner
cognitive
loop
and
then
one
other
note
there
with
this
last
section
of
what
you
wrote
is
when
we're
using
the
active
block
for
package
which
is

implementing

active

inference

in

CAD

CAD

a

complex

systems

modeling

framework

CAD

CAD

enables

parameter

sweeps

so you

can have

a parameter

inside your

simulation

0.1

and then

you can

test

and run

a bunch

of

simulations

concurrently

or in

series

like

what if

it were

0.1

0.2

0.3

0.4
we've
also
thought
towards
using
active
inference
exactly
in this
meta
Bayesian
way
because
the
parameter
sweep
space
can
become
vast
if you
have
hundreds
of
parameters
the
sweeping
across
hundred
dimensional
spaces
is
massive
the
space
is
large

which
actually
motivates
an
imperative
for
searches
and that
brings
timeliness
back
into
the
picture
through
the
scientific
modeling
process
in
finite
lives
finite
resources
and so
on
where
we're
using
active
inference
and
finding
optimally
informative
experiments
of
parameters

to
test
about
the
inner
system
and so
this
is
like
a
case
where
in
the
visual
graphical
analytical
framing
shown here
in figure
nine
one
or
in the
computational
implementation
with
active
block
fronts
!
in
both
of
those
technical
settings

mathematical
and
computational
and
then
this
setting
whatever
it is
that's
on this
slide
all
of
those
are
showing
it's
not
even
just
like
left
eye
open
right
eye
open
this
is
binoculars
we're
not
just
switching
the
monocle
from

one
eye
to
another
it's
actually
able
to
be
like
wrapping
and
gripping
itself
in
this
way
Stephen
thanks
if you
go back
to that
previous
slide
that
ties
to
this
from
the
textbook
I don't
know if
you still
got it
there
the
textbook

image
yeah
the
there's
where it
says
subjective
model
I would
imagine
that itself
could be
called
might be
thought of
as an
instrument
of sorts
or a
modeling
process
there's
a question
about how
this can
be
interpreted
and
this
is
useful
maybe
this
is
a
bridge
because
even

within
this
it's
like
well
subjective
is an
interesting
term
where
it's
actually
now
being
modeled
out
where
is
it
objective
where
is
it
where
would
that
then be
in terms
of
you know
is it
a case
of
really
getting
a feel
for
these

parameters
as they
sweep
you find
the sweet
spots
you find
that
scale
friendliness
that starts
to emerge
could be
enough
I mean
that could
be what
a lot
of organisms
are doing
as well
although
it's refined
obviously
through
evolution
but
you know
as you
sweep
it is
what
pragmatically
gets you
where you
need
to
be

as
opposed
to
being
real
values
that
are
being
aimed
for
so you
get
into
this
interesting
loop
I think
that
Dean
was
showing
there
between
the
realism
anti-realism
I mean
there might
even be
times to
break
realism
to get
out of
that
hard
lock

and say
this
is
really
what's
going
on
no
it's
a
sweep
to
find
a
scale
friendliness
it's
you know
Dean
was
going to
say
!
without
any
sort
of
prompting
that
diagram
that
I
put
up
I
don't
know
when

Carl
built
that
diagram
I
know
when
I
built
mine
and
then
ported
it
over
to
this
set
of
slides
it
was
many
many
years
ago
and
the
thing
that
I
always
said
to
people
who
I
shared

it
with
and
I
tried
to
unpack
it
and
explain
it
was
I
want
you
to
manipulate
it
well
the
capacity
as
host
and
moderator
of
these
live
streams
he
takes
information
and
he
manipulates
it
which
is

exactly
what
I
had
participants
in
the
programming
that
I
was
doing
actually
do
because
if
it's
left
as
a
representation
and
the
person
isn't
allowed
to
move
things
around
and
actually
play
with
it
then
they
have

to
take
they
have
to
they
I
I'm
imposing
on
them
where
the
person
reading
Carl
books
and
Carl
representation
that
process
to
product
like
there's
a
product
now
in
front
of
you
that's
already
been
established
and

now
you're
interpreting
that
but
every
time
you
manipulate
it
you
actually
start
by
creating
your
own
product
first
and
then
through
the
manipulation
discover
the
relationship
and
the
relativity
so
one
of
the
things
whether
it's
Carl's

representation

or

my

much

cruder

one

it's

still

when

the

person

picks

it

up

and

literally

sweeps

through

it

Stephen

as you

were

describing

it

that's

when

the

inversion

actually

takes

place

if

you

leave

it

static

if

you

leave
it
as
a
bridge
and
then
you
don't
get
a
chance
to
move
it
around
and
manipulate
it
and
re-represent
it
through
your
own
inaction
that's
when
we
tend
to
get
I
some
of
!
are
!

super
bright
I'm
not
one
of
those
people
that's
why
I
had
to
give
people
like
me
the
opportunity
to
say
don't
just
leave
it
on
the
page
play
with
it
do
something
with
it
enact
it
into

your
own
set
of
comprehensions
so
here's
a
little
evocative
image
meme
but
let's
to
complete
the
comparison
we
discussed
how
we
had
the
open
edge
quadrant
model
here
and
some
quotes
from
the
inactive
dynamic
paper
now

we're
going
to
go
to
the
next
slide
where
the
bounding
of
the
red
is
on
the
outside
of
the
visual
and
we're
taking
quotes
from
the
abducted
paper
so
what
quotes
did
you
highlight
from
the
paper

and
what
does
the
change
in
the
visualization
represent
okay
so
first
thing
they
see
is
that
the
rows
and
the
columns
have
disappeared
and
all
we've
done
is
made
sure
that
there's
those
edges
on
the
outside

so
just
to
confirm
that
so
remember
this
is
just
comparison
one
there
there
were
six
comparisons
between
these
two
papers
that
I
pulled
so
I
want
to
make
that
clear
too
but
on
this
first
comparison
just

to
show
what
the
difference
is
between
an
edge
list
and
an
edge
constrained
this
is
especially
the
case
this is
a quote
now
from
Adi
and
Majid's
paper
this
is
especially
the
case
that
we
assume
that
the
development

of
markup
blankets
by
Pristen
now
eclat
Pristen
blankets
unlike
their
original
characterization
works
by
Judea
Pearl
on
probabilistic
causal
networks
admit
of a
realist
interpretation
of the
relationship
between
the
organism
and
the
environment
while
the
role
so
it's

variability
retained
the role
of the
concepts
of the
Markov
blanket
and
Pro
blankets
and
many
treatments
of
the
free
energy
principle
is
an
instrumentalist
one
whereas
in
Pierce's
even
stronger
terms
concede
to
a
nominal
interpretation
of
meaning
Pristen
blankets

are
realist
not
only
in
the
non
instrumental
sense
but
also
in
the
sense
of
being
non
nominal
entities
the
modal
terms
of
references
are
two
real
possibilities
such
as
what's
conceivable
or
produced
by
imagination
anticipatorially
related to

the future
states
of affairs
are
used
in
making
sense
of
the
key
aspects
of
the
theory
we
will
argue
in
this
paper
the
free
energy
principle
has
been
put
at
the
center
of
a
viable
picture
of
an
organism

insight
into
the
consequences
of
its
action
and
its
ability
to
choose
the
best
or
most
efficient
strategies
that
contribute
to
the
fulfillment
of
best
possible
consequences
so
not
just
consequences
now
but
over
a
temporal
field
and

all I
noted
was
realism
is
the
philosophical
position
that
posits
that
universals
are
just
as
real
as
physical
measurable
material
nominalism
is
the
philosophical
position
that
promotes

I
just
lost
my
place
sorry
I
just
made
a

new
line
abstract
concepts
do
not
exist
in
the
same
way
as
physical
tangible
material
from
instrumentalism
to
insight
and
insight
or
minimum
to
we
talk
about
that
all
the
time
because
we're
talking
about
relational
and
not

just
relative
to
rule
we
build
categories
we
generate
hierarchies
and
eventually
we
see
patterns
to
model
we
gain
an
insight
then
the
pattern
is
tested
from
Adi
there's
another
paper
that I
read
that's
fantastic
called
the
new

light
on
pierce
and
I
quoted
from
that
this
is
what
pierce
had
in
mind
when
he
wrote
lady
well
be
that's
reasonable
to
adopt
a
given
hypothesis
reasonable
right
if
it
will
be
wise
to
go
to

some
expense
to
put
the
hypothesis
to
test
dependent
upon
the
advantage
that
that
would
accrue
from
knowing
the
truth
of
the
hypothesis
from
that
testing
result
of
consistency
i.e.
repeatability
comes
universal
things
like
laws
rules
axioms

etc
that's
the
stuff
that
in
my
diagram
is
on
the
right
hand
side
this
is
the
entirety
of
the
process
real
yes
because
it's
complex
a
reality
made
up
of
processes
some
of
which
have
no
basis

in
reality
again
from
the
new
light
on
pierce
again
adi's
paper
the
fourth
memoir
entitled
analysis
of
the
methods
of
mathematical
demonstration
opened
with the
affirmation
that
his
first
real
discovery
about
mathematical
procedure
was
that
there
are

two
kinds
of
necessary
reasoning
the
collateral
and
the
theoret
theoret
theoret
these
he
explained
as
follows
collateral
deduction
is
where
it
is
only
necessary
to
imagine
any
case
in
which
the
premises
are
true
in
order
to

perceive
immediately
that
the
conclusion
holds
in
that
case
the
aromatic
deduction
is
deduction
in
which
it's
necessary
to
experiment
in
the
imagination
upon
the
image
of
the
premise
then
in
order
from
the
result
of
which
experiment

to
make
sorry
collateral
deductions
to the
truth
of the
conclusion
so
again
this

hints
that
if
we
remove
too
many
things
and
reduce
too
far
we're
left
with
nothing
but
instrumentalism
so
within
those
bounded
contexts
!
that

we've
now
arbitrarily
parameterized
there's a whole bunch
of things
that we can
pack into
that
silo
and those
things can
move around
and float
around in
there
they're
not
constrained
to the
upper
right
and the
lower
left
corner
now
all of a
sudden
those
bidirectionalities
are
indicators
of just
how much
fluidity
is going
on as

we pass
from
product
and
sweep
across
to
insight
good
thanks
dean
let me
add
a thought
here
so
just
for
reminder
and
reference
other
resources
you'll
find
so
much
more
but
deduction
is
the
inference
of
or
about
particular
instances

by
referencing
generalities
laws
and
principles
induction
in
contrast
is
where
we
go
from
what
we
observed
with
the
instances
up
towards
forming
generalities
so
deduction
going
down
in that
kind
of a
top
down
framing
it's
very
interesting
that

the
deduction
is
split
into
two
components
here
you
said
collateral
I
believe
and I
think
that
may
be
totally
valid
but
I
at least
would
read
this
as
corollary
as
in
of
or
related
to
a
corollary
well
what

have
we
heard
nowhere
else
but
here
and
elsewhere
octave
inference
is a
corollary
process
theory
of
the
free
energy
principle
so
from
a
more
general
principle
in the
definition
the
corollary
deduction
is
a
situation
where
a
case
is

only
needed
to
be
imagined
really
imagined
where
the
premises
are
true
propose
a
blanket
where
there's
a
particular
entity
and
the
blanket
is
separating
them
from
the
generative
process
in
such
and
such
a
way
such
that

the
entity
is
a
generative
model
and
the
internal
states
are
doing
some
kind
of
mode
tracking
of
external
states
in a
cybernetic
good
regulator
framework
the
premise
is
merely
a
premise
it's
not
the
culmination
so
that's
what

we
see
in
the
corollary
deduction
all
you
have
to
do
is
consider
the
premise
to
perceive
immediately
that
the
conclusion
holds
in
that
case
well
yes
if
you
propose
setting
up
the
base
graph
to
be
truly

just
that
way
as
a
corollary
of
the
free
energy
principle
then
that
is
already
like
a
story
arc
that's
been
completed
no
one's
saying
that
you
wrote
the
paper
or
that
you
had
the
startup
that's
profitable

but
also
no
one
can
say
that
it's
incorrect
that
you
made
a
premise
that
kind
of
completed
its
arc
and
then
in
the
second
case
we're
still
within
deduction
however
it's
necessary
to
experiment
in
the
imagination

so
a
thought
experiment
which
we
can
place
into
a
setting
of
mental
action
and
attention
and
counterfactuals
counterfactual
generation
requires
the
generation
of
counterfactuals
ergo
it is
abductive
it requires
a generative
phase
and a
selective
phase
so
we're
going to
generate

and
select
that's
the
abductive
loop
that's
happening
with a
when
in a
semantic
space
as
part
of
this
theorematic
deduction
and
then
the
results
of
that
experiment
can
be
used
to
make
corollarial
deductions
to the
truth
of
the
conclusion

so
that's
kind
of
counterfactuals
on
imagined
premises
and
guess
what
it's
real
Stephen
can
I add
one
question
in
there
we
have
our
make
our
deduction
from
a
particular
instance
and
we
abduct
the
kind
of
best
policy

selection
when
we
induce
generally
this
induction
I
see that
as
kind
of
a
narrowing
the
gap
often
like
there's
a
generality
and then
a bit
like
reinforcement
learning
like
trying
to
you're
inducing
between
and
that's
often
what we
see
is

most
visible
you know
someone's
trying to
reduce
the gap
in the
physical
world
with the
product
that we've
got
that we're
manipulating
because that's
kind of what
we can see
I was speaking
I think
earlier to
Dean about
this
you know
someone's
making a
pot
and that
you know
they have
the piece
of clay
that goes
wobbling
and they're
sort of
making it

into a
pot
you know
so there's
kind of
like
they're
inducing
inductively
trying to
make it
closer and
closer to
something
that's
more like
the general
idea of
a pot
but
that's
what we
see
and we
at some
point we
might not
know what
they're even
doing
then suddenly
we deduce
it's a
pot
because like
you say
there's a
particular

instance
of being
evoked
but behind
all of
that
there's
abducted
processes
the
biosem
I don't
know
whether
the word
biosemiotics
is the
term
actually
I'm still
not quite
sure how
to put
that in
there
but
I think
that fits
in with
what you're
saying
I wonder
what your
thoughts
are on
that sort
of
inductive

take
if I can
just
highlight
how this
comparison
one
relates to
the realism
instrumentalism
and the
clay pot
creation
so here
we have
the quadrant
separated
again
totally
speculating
working

with
Dean's
graphical
artistry
in
one
totally
defensible
and
absolutely
scientific
framing
the viewer
of the
pot
construction

could see
their entire
logical
apparatus
and inference
about what
is happening
as an
instrumental
process
happening
I'm not
saying
they're
really
making
a
pot
I'm
just
saying
that's
a
socially
constructed
term
and it's
totally
contingent
on my
enculturation
what I
believe is
a pot
maybe they're
making a
piece of
art

who am I
to say
that it's
not something
else
maybe it
could be
used as
a hammer
so
that
places
the
inference
about
the
system
at
the
forefront
and
provides
clarity
as well
as
openness
the edges
of the
drawing
are
open
because
it does
leave a
lot of
structuredness
surrounding
the

origin
point
again
but also
room to
go off
in many
many
directions
what
Majeed
returned
our
attention
to
was
the
nature
of
observations
coming from
the generative
process
coming from
the niche
if it's
whimsical
as I
think he
said
well then
yes
instrumentalism
is fine
in its
totality
however if
it's not

whimsical
if there
really is
a table
that you're
going to
get hit
by
if that
is really
something
that you're
buying
if that
is something
with limited
resources
we're doing
are we
making a
pot
or are we
making a
pan
then
the
boundedness
is now
moving to
that
scenario
which includes
the
the
mindedness
of all
of the
entities

one entity
is not
speaking for
the other
but that
entity
is seeing
the situation
as real
in that it
is truly
providing
pragmatic
implications
for survival
and
its real
meaning

it's really
biosemiotic
meaning
and so
here
the
boundedness
is
more
around
the
situation
and
in
that
space
in that
room
there

are
paths
amongst
the
theory
where
if you
go into
a different
situation
you cross
the line
now
crossing
a line
is not
like
always
a bad
thing
but
all
of a
sudden
if
someone's
daydreaming
about
a totally
different
topic
they've
exited
the
situation
but
within
the

situation
there
are
diversity
of
approaches
here
the
situation
is
actually
being
used
just
as
an
experimental
context
for a
theoretician
or a
researcher
in their
removal
to be
able to
consider
different
ways of
modeling
so
I'm not
sure if
this is
consistent
or
you know
dean

orthodoxy
or
heterodoxy
but I
see the
boundedness
as
allowing
for
that
center
of
action
to be
approached
in a
pragmatic
not
a
theoretical
but not
academic
approach
whereas
by making
the divisions
cutting nature
at the
joints
and like
oh wow
those joints
correspond to
long-standing
philosophical
debates
in academia
who knew

and maybe
they don't
always
correspond
to those
joints
yes
dean
I know
we're getting
we're getting
close to the
end here
so thank
you for
explaining that
in your
words
that helps
me because
I'm always
curious to
whether or
not the
things that
I'm trying
to share
actually are
making sense
to other
people
and it
sounds like
it maybe
is is
gaining
some traction
I want

to say
this one
thing
before we
have to
sign off
here
traditionally
in most
academic
settings
and I'm
as guilty
of this
as anybody
because I
was a
curriculum
designer
for a part
of my
life
we start
out
trying to
figure out
how to
construct a
plan
and I
think
what
active
instrument
affords
is this
idea
that

instead
of
instead
of
reconstruct
a plan
and then
others
follow it
called a
curriculum
what this
potentiates
in addition
to that
this is not
dismissing
that
or saying
that that
doesn't have
its time
and its
place
what this
does is
asks us
whether or
not we
could
forage
a plan
could we
figure out
how to
fly
how to
build the

plane
on the
fly
that's
all this
is
entertaining
so that
you actually
have both
so when
you're
a sully
and Hudson
River
has never
traditionally
been a
landing strip
it becomes
one
but you
need both
you need
the think
and the do
you need
the action
and the
abduction
and so
if you
if you
limit yourself
to one
or the
other
for the

sake of
not being
cognitively
overloaded
I get
that
that's
the
deconstruction
piece
but if
you are
foraging
for a
plan
the plan
falls
out
of a
product
your
priors
meld
with your
priorities
and then
it becomes
self-evident
what the
plan is
as opposed
to
somebody
handing
you
a list
and then
you simply

following
so I
wanted to
get that
in before
we
have to
call it
a day
but I
think
that's
important
to
put
that
context
both
in
gripped
and
as
gripper
because
it's
no more
complicated
or simple
than that
we
are
on our
last slide
now
we got
through
one of
the six

comparisons
and the six
were the tip
of the iceberg
of the tip
so
we know
that we can
never
linearize
or
vocalize
everything
that needs
to occur
but
we also
carry on
so
we can
pause
here
reflect
on
the last
three
videos
in this
series
47
0
1
and
2
where
to
now
how

are we
different
after
47.2
than
before
47.00
Dean
my suggestion
is that
when we
reflect
we are
literally
abducting
we are
literally
looking back
at the
entire
narrative
or
continuum
or
whatever
you want
to call
it
and trying
to figure
out
what just
happened
so
yeah
and I
think
that's

when
a lot
of the
stuff
that's
that's
non
instrumentalist
turns
real
I don't
think
people
actually
get a
real
sense
of
real
until
they
do
the
reflection
the
classical
scientific
communication
public
speaking
advice
which
I
look
forward
to
seeing
how

what
Majid
shared
about
economy
of
research
also
reimagines
the
doing
and the
communication
of science
of course
but
some
of the
classic
advice
is
tell them
what you're
going to
tell them
tell them
and then
tell them
what you've
told them
we can
turn that
into a
bit more
of a
inactive
or a
planning

based
framing
like
consider
what is
about to
happen
abduction
during the
live stream
think about
what is
happening
attention
and then
after the
live stream
consider
what happens
reflection
all of
those
processes
are
happening
in the
.01
and 2
of each
sentence
of each
person
speaking
turn
in the
videos
and at
all these

different
scales
that's as
real
as anything
so it's
really
a nice
hopefully
sequence
that we've
engaged
on
and there
may be
more decimals
to come
on 47
certainly
48 and
beyond will
also carry
forward in
this
historical
way
in which
when we
talk about
concepts
beyond 47
we'll have
this shared
basis
of bringing
abduction
in and
so on

so
Stephen
any
last
thoughts
yeah
I'd like
to
thanks for
going
here
I think
we've had
enough
traction
to show
that there's
a lot
of value
in
trying
to
wrestle
between
and
there
is
as well
as that
idea
of the
scientific
you know
like
Daniel
you mentioned
the scientific
communication

where you
have
something
that's
quite
clear
to
communicate
there's
also
when it's
opening up
a space
for
something
that could
go down
multiple
different
routes
once it's
contextualized
so I think
this speaks
to
contextuality
situatedness
and may
require
some
I'm
interested
in how
this may
require
some
nonverbal
kind of

more
experience
experiential
processing
to
like
of
certain
cases
to
explore
so I
think
that
might
be
open
up
the
potential
to
sit
in
between
and
just
take
something
from
here
and say
okay
what
would
that
be
like
experientially

and just
groove in
that
so that's
something
I'm
going to
explore
what
that
might
be
like
Dean
penultimate
words
I
want
to
thank
the
two
of
you
again
for
creating
a
kind
of
safe
space
to
take
these
risks
because
again

we're
not
just
sticking
to
the
script
we're
maybe
triangulating
off
two
scripts
and
that
doesn't
mean
that
you're
always
going
to
be
right
I
mean
if
people
come
back
and
say
wow
are you
wrong
about
this
I

get
to
learn
but
I
think
before
that
opportunity
arose
prior to
that
there
were
two
people
that
are
now
sitting
on
my
desktop
here
who
shared
in
that
risk
analysis
effort
and I
want
to
thank
both
you
for

being
so
supportive
around
taking
this
kind
of
let's
hang
it
out
there
and
see
who
salutes
and
who
takes
it
down
because
if
it's
taken
down
I'm
okay
with
that
too
but
just
being
able
to
have

the
opportunity
has
been
spectacular
in
terms
of
developing
how
I
think
ask
not
what
you
can
do
for
your
corollary
process
theory
see you
fellows
later
thank
you
thank
you
thank

Thank you.